Asian College of Higher Studies

Ekantakuna, Lalitpur

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Lab Report

Department : Computer Science & Information Technology
Course Title : Software Project Management
Course No : CSC 425
Experiment No : 02
Experiment Name: Construction of Gantt Chart, Bubble chart & Slip Line chart

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Lab 2: Construction of Gantt Chart, Slip chart & Bubble chart

Objectives:

a) To understand project scheduling techniques
b) To construct a Gantt chart for project planning.
c) To analyze project delay using slip line chart
d) To represent cost, duration & risk using Bubble chart
e) To use MS Excel for creating professional project charts.

Project Scenario:

Development of an online Student Admission Management System for a university. The system includes registration, document upload, online payment, merit list generation, & admin reporting. Some activities can be executed in parallel to reduce project duration.

a) Design & construct a Gantt chart showing overlapping (parallel) activities in project schedule.

b) Design & construct a slip line chart comparing planned versus actual progress of parallel activities

c) Design & construct a Bubble chart to represent cost, risk & duration of parallel project activities.

Introduction:

Project Management tools are essential for planning, organizing & controlling software development projects. In this lab, project management charts are constructed for the development of an Online Admission Management system. The system includes student registration, document upload, payment integration, merit lists generation & reporting modules.

Theory:

a) Gantt Chart:

A Gantt chart is a bar chart used to represent project activities against time. It shows start time, end time, duration & parallel execution of tasks

- Advantages:
  - Easy visualization of schedule
  - Shows overlapping activities
  - Useful for monitoring progress

- Disadvantages:
  - Does not clearly show task dependencies
  - Difficult for very large project.

b) Slip Line Chart

A slip line chart compares planned progress with actual progress over time. It helps identify schedule delays & performance gaps.

- Advantages:
  - Detects project delay
  - Simple to interpret
- Disadvantages
  - Requires continous updating
  - Limited activity - level detail.

c) Bubble chart:

A Bubble chart displays three variables simultaneously

- X-axis -> Duration
- Y-axis -> Cost
- Bubble Size -> Risk level

It helps analyzing cost-time-risk relationship.

<u>Project Activities:</u>

Project Duration: 16 Weeks

ID	Activity	Duration	Start	End
A1	Project Initation	1	1	1
A2	Stakeholder Meeting	1	1	1
A3	Requirement Gathering	2	2	3
A4	Feasibility Study	1	2	2
A5	SRS Documentation (Software Requirements Specification)	1	3	3

A6	System Architecture Design	2	4	5
A7	Database Design	2	4	5
A8	UI/UX Design	2	4	5
A9	Frontend Development	3	6	8
A10	Backend Development	3	6	8
A11	Student Registration Module	2	9	10
A12	Document Upload Module	2	9	10
A13	Payment Gateway Integration	2	9	10
A14	Merit List Generation	2	11	12
A15	Notification System	2	11	12
A16	Admin Dashboard	2	11	12
A17	Integration Testing	2	13	14
A18	System Testing	2	13	14
A19	User Acceptance Testing	1	15	15
A20	Deployment & Trainning	1	16	16

Gantt chart:

Title:

Gantt chart for online Student Admission Management System Project Schedule:

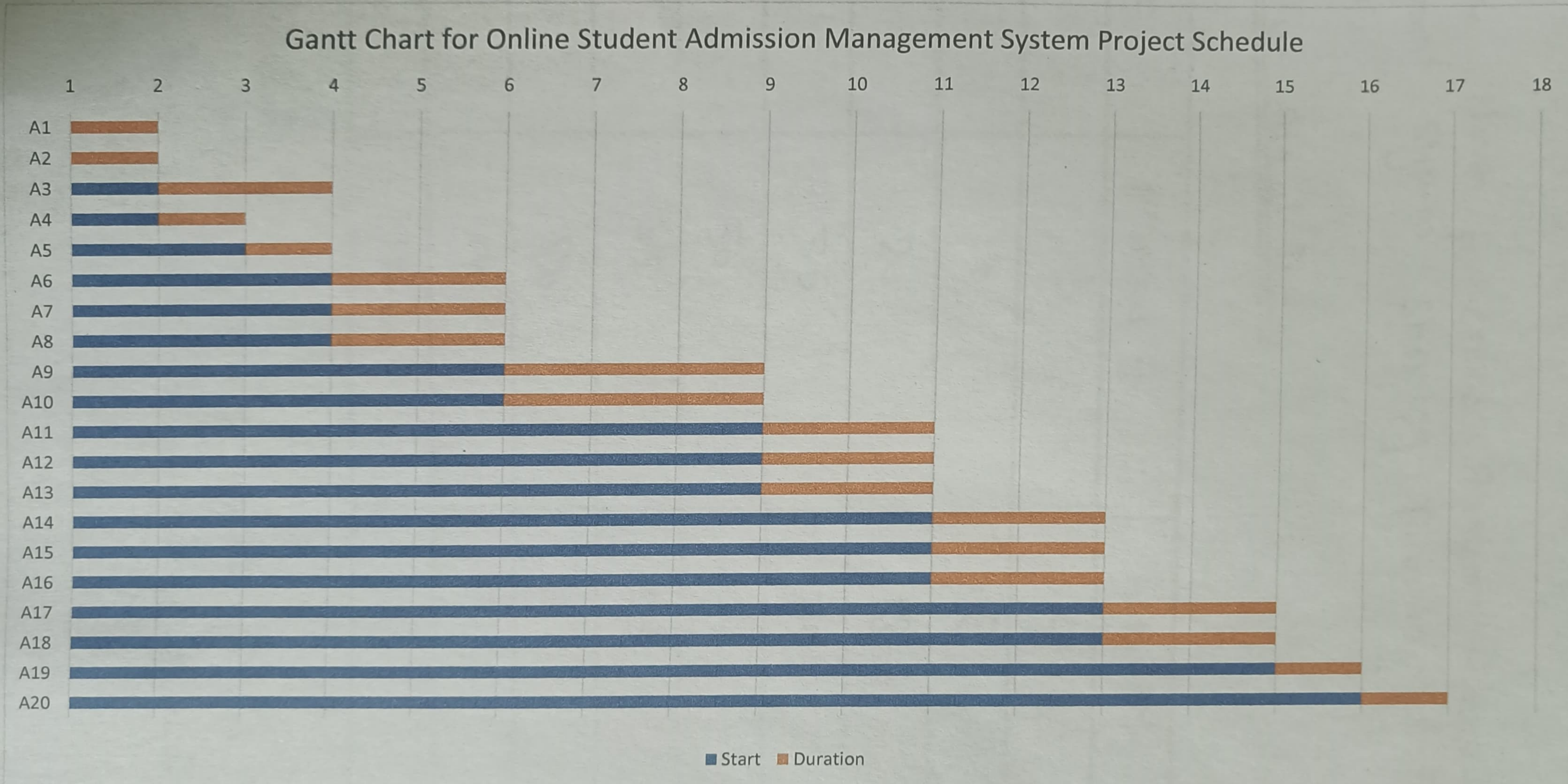
The chart shows project Schedule with 20 activities & parallel execution.

Parallel Activities:

- A1 & A2
- A3 & A4
- A6, A7 & A8
- A9 & A10
- A11, A12 & A13
- A14, A15, & A16
- A17 & A18

Interpretation:

The chart clearly represents overlapping activities which reduces total project duration.


Gantt Chart for Online Student Admission Management System Project Schedule
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
A16
A17
A18
A19
A20
Start
Duration

Slip Line Chart:

Title:

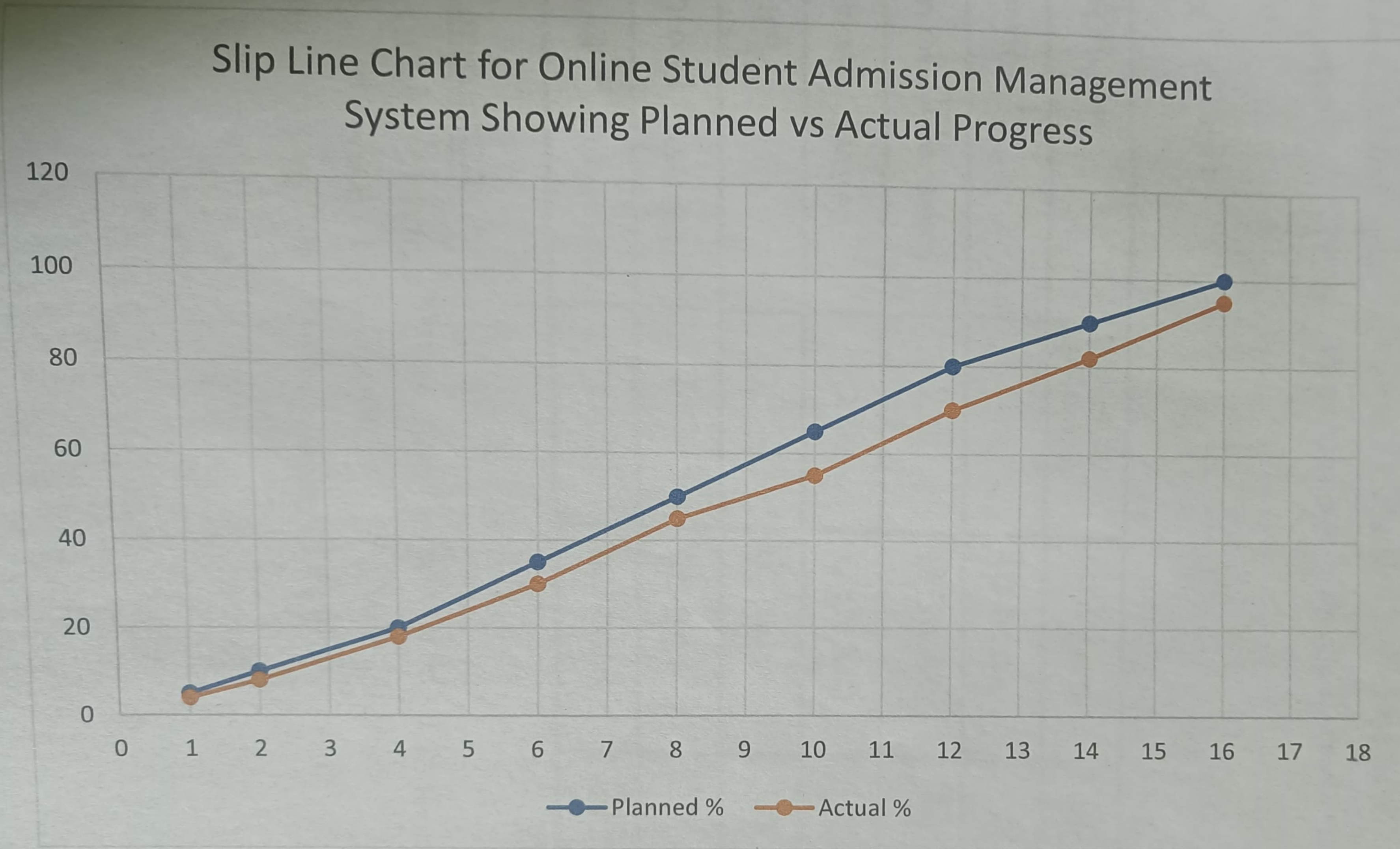
Slip Line chart for online Student Admission Management System showing Planned vs Actual Progress

Planned vs Actual Data:

Week	Planned %	Actual %
1	5	4
2	10	8
4	20	18
6	35	30
8	50	45
10	65	55
12	80	70
14	90	82
16	100	95

Interpretations:

The actual progress is slightly below planned progress indicating minor schedule delay during development & testing phases.



Slip Line Chart for Online Student Admission Management System Showing Planned vs Actual Progress
120
100
80
60
40
20
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Planned %
Actual %

Bubble chart:

Title:

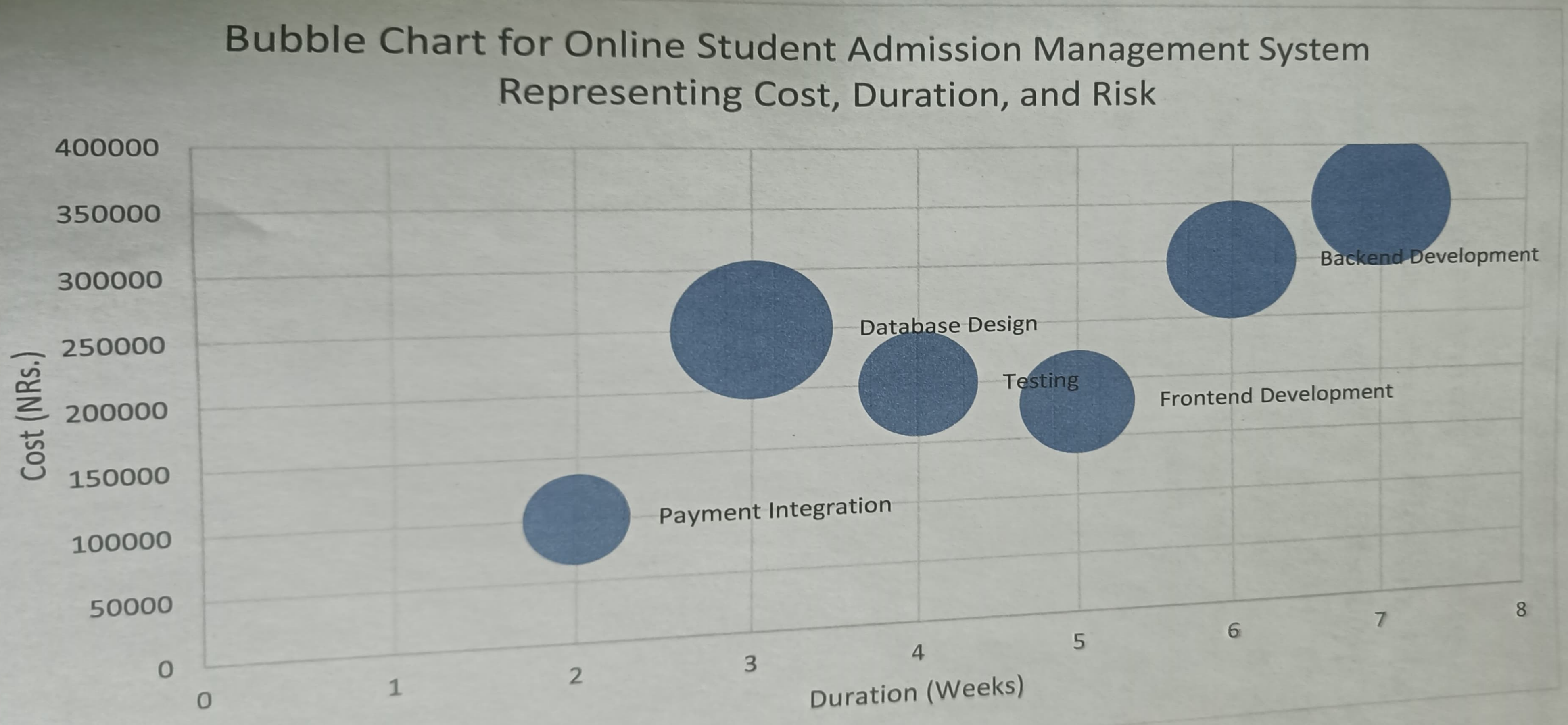
Bubble chart for Online Student Admission Management System Representing Cost, Duration & Risk.

Data used:

Activity	Duration (Weeks)	Cost (Npr)	Risk (1-5)
Deployment	1	10000	2
Payment Integration	2	250000	5
Database Design	2	200000	3
Testing	2	180000	3
Frontend Development	3	300000	4
Backend Development	3	350000	5

Interpretation:

The bubble chart shows that Backend Development has the heighest cost, duration & risk, while Paymant integration have heighest risk.


Bubble Chart for Online Student Admission Management System
Representing Cost, Duration, and Risk
Cost (NRs.)
400000
350000
300000
250000
200000
150000
100000
50000
0
0
1
2
3
4
5
6
7
8
Duration (Weeks)
Payment Integration
Database Design
Testing
Frontend Development
Backend Development

Conclusion:

[redacted] .

In this lab project management charts were constructed for the online student Admission Management system. The Gantt chart helped in scheduling & visualizing parallel activities. The slip line chart compared planned & actual progress to detect delay. The Bubble chart analyzed cost, duration & risk of major [redacted] activities. These tools are essential & effective for project planning & control.